Indian Institute of Technology Guwahati
Department of Electronics and Electrical Engineering
EE101 - Basic Electronics

Tutorial - 10 **Date: Nov 2, 2023**

1. Determine the dc bias voltage V_{CE}, the current I_C and the ac emitter resistance r_e for the voltage-divider configuration given in Fig. 1. Let $V_{CC} = +20V$, $\beta = 100$, $R_1 = 42k\Omega$, $R_2 = 4.2k\Omega$, $R_C = 12k\Omega$, $R_E = 1.5k\Omega$.

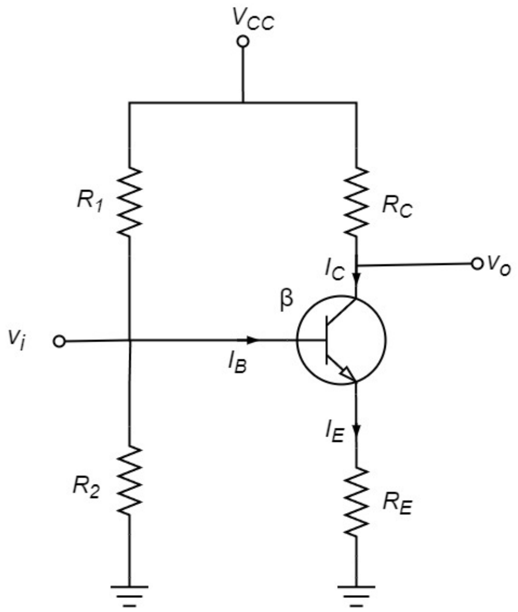

Figure 1

2. Determine the value of I_C, V_{CE} and ac emitter resistance r_e for the network of Fig. 2.Let $V_{CC} = +12V$, $\beta = 100$, $R_B = 250k\Omega$, $R_C = 5k\Omega$, $R_E = 1.2k\Omega$.

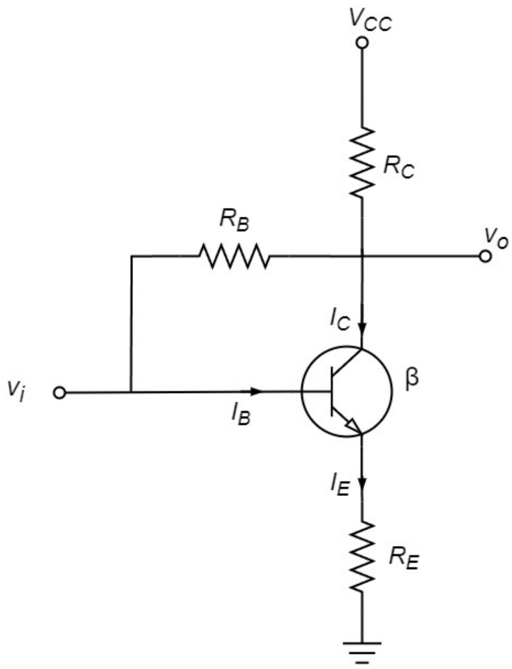

Figure 2

3. Refer to the circuit in the figure below. Determine the total average power, reactive power, and complex power at the source and at the load.

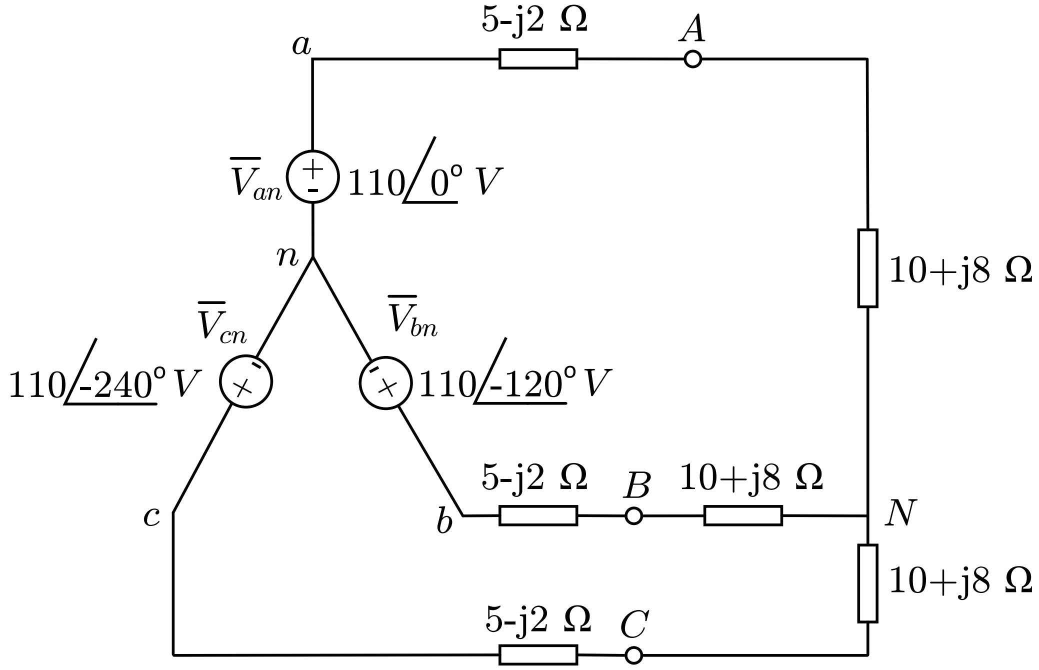

Figure 3 Three-wire Y-Y system.

4. Two balanced loads are connected to a 240 *kV*, 60 *Hz* line, as shown in Figure below. Load 1 draws 30 *kW* at a power factor of 0.6 lagging, while load 2 draws 45 *kVAR* at a power factor of 0.8 lagging. Assuming the *abc* sequence, determine:
   (A) the complex, real, and reactive powers of the combined load
   (B) the line currents
   (C) the *kVAR* rating of the three capacitors Δ-connected in parallel with the load that will raise the power factor to 0.9 lagging and the capacitance of each capacitor.

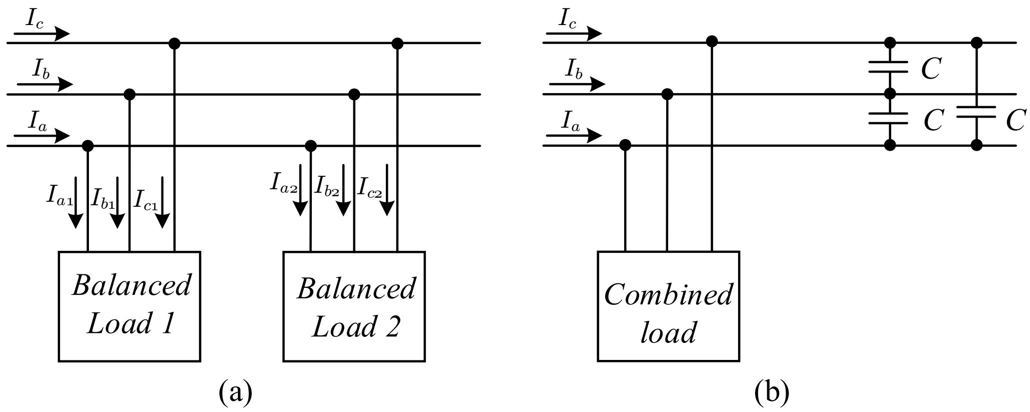

Figure 4 (a) The originally balanced loads, (b) the combined load with improved power factor.